

AWS OWNS LAB

Design and Implementation of a SOC Lab using AWS EC2

INTRODUCTION.

A Security Operations Center (SOC) is a centralized unit responsible for monitoring, detecting, analyzing, and responding to cyber security incidents in real time. With the rapid growth of cyber threats, organizations require effective monitoring systems to ensure the security of their infrastructure.

This project focuses on designing and implementing a SOC lab environment using cloud computing technology. The lab is built using **Amazon Web Services AWS**), specifically leveraging **Amazon EC2**, to simulate real-world security monitoring operations. A Security Information and Event Management (SIEM) tool, **Splunk**, is used to collect, analyze, and visualize log data.

OBJECTIVES.

The main objectives of this project are:

- To design and deploy a SOC lab using cloud infrastructure
- To configure and manage a virtual server using **Amazon EC2**
- To install and configure a SIEM tool **Splunk**)
- To simulate log collection and monitoring processes
- To gain practical experience in cybersecurity operations

System Architecture

The SOC lab environment consists of multiple components deployed in the cloud:

- **Splunk Server (EC2 Instance)** – Used for log collection, indexing, and analysis
- **Windows Server (EC2 Instance)** – Configured with Active Directory as a log source
- **Client Machine (Kali Linux)** – Used for accessing and managing the environment
- **Splunk Universal Forwarder** – Installed on the Windows Server to send logs to Splunk

All components communicate over the internet. Logs generated from the Windows Server are forwarded to the Splunk server, where they are analyzed and visualized through a web interface.

Tools and Technologies

The following tools and technologies were used:

- **Amazon Web Services (AWS)** – Cloud infrastructure
- **Amazon EC2** – Virtual servers
- **Splunk** – SIEM platform
- Windows Server – Active Directory configuration
- Kali Linux – Client machine
- SSH – Remote access to Linux server
- RDP (Remmina) – Remote access to Windows server

Launching EC2 Instance

An EC2 instance was created using the AWS management console. A Linux-based operating system was selected, and a key pair was generated for secure access. Security group rules were configured to allow inbound traffic on required ports such as SSH port 22) and Splunk web interface port 8000).

So in order to do all those this your need first to have an account of AWS and then continue to AWS EC2 instance and then navigate to launch instance. For example the pictures below show this process.

Launch an instance [Info](#)

Amazon EC2 allows you to create virtual machines, or instances, that run on the AWS Cloud. Quickly get started by following the simple steps below.

Name and tags [Info](#)

Name

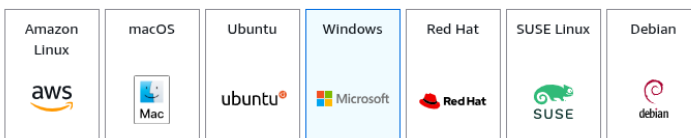
[Add additional tags](#)

▼ Application and OS Images (Amazon Machine Image) [Info](#)

An AMI contains the operating system, application server, and applications for your instance. If you don't see a suitable AMI below, use the search field or choose [Browse more AMIs](#).

Recents

Quick Start



[Browse more AMIs](#)

Including AMIs from
AWS, Marketplace and
the Community

Amazon Machine Image (AMI)

Microsoft Windows Server 2025 Base
ami-0798c9ebaf695e14e (64-bit (x86))
Virtualization: hvm ENA enabled: true Root device type: ebs

Free tier eligible ▼

so the diagram show the launch instance name and a place to select the operating system ,so the picture show my window server for creating the active directory and below are for splunk.

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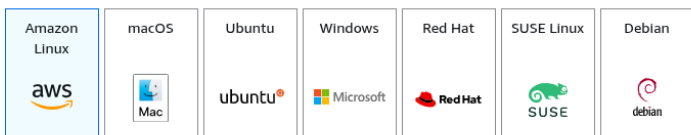
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Amazon Machine Image (AMI)

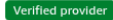
Amazon Linux 2023 kernel-6.1 AMI
ami-098e39bafa7e7303d (64-bit (x86), uefi-preferred) / ami-03ed25db53d8de46c (64-bit (Arm), uefi)
Virtualization: hvm ENA enabled: true Root device type: ebs

Free tier eligible ▼

After that its followed by choosing the instance type which contain the memory(RAM) ,numbers of CPU and many other details.

Description
Amazon Linux 2023 (kernel-6.1) is a modern, general purpose Linux-based OS that comes with 5 years of long term support. It is optimized for AWS and designed to provide a secure, stable and high-performance execution environment to develop and run your cloud applications.

Amazon Linux 2023 AMI 2023.11.20260413.0 x86_64 HVM kernel-6.1

Architecture	Boot mode	AMI ID	Publish Date	Username	
64-bit (x86)	uefi-preferred	ami-098e39bafa7e7303d	2026-04-10	ec2-user	

▼ Instance type [Info](#) | [Get advice](#)

Instance type
t3.small Free tier eligible
Family: t3 2 vCPU 2 GiB Memory Current generation: true On-Demand RHEL base pricing: 0.0496 USD per Hour
On-Demand Windows base pricing: 0.0392 USD per Hour On-Demand SUSE base pricing: 0.0518 USD per Hour
On-Demand Linux base pricing: 0.0208 USD per Hour On-Demand Ubuntu Pro base pricing: 0.0243 USD per Hour

☐ All generations
[Compare instance types](#)

[Additional costs apply for AMIs with pre-installed software](#)

▼ Key pair (login) [Info](#)

You can use a key pair to securely connect to your instance. Ensure that you have access to the selected key pair before you launch the instance.

Key pair name - required
splunklab 

 [Create new key pair](#)

After this step is followed by creating a key pair which will give us a password for login in remote in our server through SSH and RDP as show below after completing selecting the file will be downloaded with the extension of .pem

which will help your to get password through decrypting the RSA so it's a key

Create key pair

Key pair name

Key pairs allow you to connect to your instance securely.

splunklab

The name can include up to 255 ASCII characters. It can't include leading or trailing spaces.

Key pair type

☒ **RSA**

RSA encrypted private and public key pair

☐ **ED25519**

ED25519 encrypted private and public key pair

Private key file format

☒ **.pem**

For use with OpenSSH

☐ **.ppk**

For use with PuTTY

 When prompted, store the private key in a secure and accessible location on your computer. **You will need it later to connect to your instance.** [Learn more](#) 

Cancel

Create key pair

After this is followed by network setting

▼ Network settings [Info](#)

Edit

Network | [Info](#)
vpc-04d96053b1bab743e

Subnet | [Info](#)
No preference (Default subnet in any availability zone)

Auto-assign public IP | [Info](#)
Enable

Firewall (security groups) | [Info](#)
A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to reach your instance.

☒ Create security group ☐ Select existing security group

We'll create a new security group called 'launch-wizard-5' with the following rules:

☒ Allow SSH traffic from
Helps you connect to your instance
Anywhere
0.0.0.0/0

☐ Allow HTTPS traffic from the internet
To set up an endpoint, for example when creating a web server

☐ Allow HTTP traffic from the internet
To set up an endpoint, for example when creating a web server

⚠ Rules with source of 0.0.0.0/0 allow all IP addresses to access your instance. We recommend setting security group rules to allow access from known IP addresses only.

After this is followed by storage configurations.

so in here a user will choose the storage in which he/she want its server to have but for those who use free there is a limited size of storage if your need more you should pay for it.

▼ Configure storage [Info](#)

Advanced

1x 8 GiB gp3 Root volume, 3000 IOPS, Not encrypted

Add new volume

🕒 Click refresh to view backup information

The tags that you assign determine whether the instance will be backed up by any Data Lifecycle Manager policies.

File systems

☐ S3 Files - new ☐ EFS ☐ FSx

After that this is the last step where a user can see the summary of all important steps he/she had done in order to get the instance where a user click to launch

instance and hence the instance will be created.

▼ Summary

Number of instances

Info

1

Software Image (AMI)

Microsoft Windows Server 2025 ...[read more](#)

ami-0798c9ebaf695e14e

Virtual server type (instance type)

t3.small

Firewall (security group)

New security group

Storage (volumes)

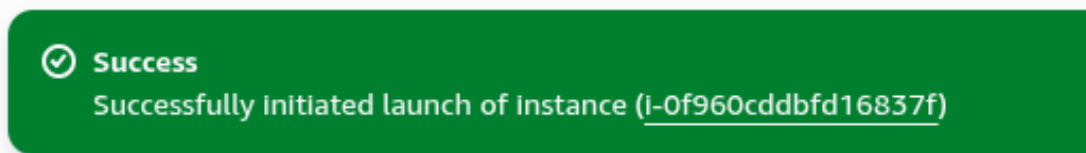
1 volume(s) - 30 GiB

Cancel

Launch instance

 [Preview code](#)

So after this the user will see this message showing that the instance is successful created.



So according to my requirement of the lab i created two instance one labeled as AD_SERVER 1 and SPLUNK LAB as show below

☐	AD_SERVER 1	i-Of960cddbdf16837f	 Running 🔍 🔗	t3.small	 Initializing View alarms +	us-east-1f	ec2-100-54-66-126.co...	100.54.66.126	-
☐	SPLUNK LAB	i-Oe03bfebaf66d831	 Running 🔍 🔗	t3.small	 3/3 checks passed View alarms +	us-east-1f	ec2-98-84-28-146.com...	98.84.28.146	-

so after the process now is followed by connecting to my server inorder to access them remotely and continue with the process of installation of splunk and active

directory on ya system in order to get my raw logs at all. The machine i used was Kali Linux OS booted so i access ya splunk lab server through ssh. in order to connect you should navigate to connect

Connect [Info](#)
Connect to an instance using the browser-based client.

Instance ID i-0e03bfeba6d6d831 (SPLUNK LAB)	VPC ID vpc-04d96053b1bab743e	Security groups sg-0f6bed1027eed2e8 (Launch-wizard-5)	IAM role -
---	--	---	----------------------

EC2 Instance Connect | SSM Session Manager | **SSH client** | EC2 serial console

Instance ID
i-0e03bfeba6d6d831 (SPLUNK LAB)

1. Open an SSH client.
2. Locate your private key file. The key used to launch this instance is splunklab.pem
3. Run this command, if necessary, to ensure your key is not publicly viewable.
`chmod 400 "splunklab.pem"`
4. Connect to your instance using its Public DNS:
`ec2-98-84-28-146.compute-1.amazonaws.com`

Example:
`ssh -i "splunklab.pem" ec2-user@ec2-98-84-28-146.compute-1.amazonaws.com`

Note: In most cases, the guessed username is correct. However, read your AMI usage instructions to check if the AMI owner has changed the default AMI username.

And for my windows server i installed a tools known as Remmina which help me to connect to my windows server through RDP. where by now the pair key you created will be used to give your the credentials inorder to access the servers remotely

Connect [Info](#)
Connect to an instance using the browser-based client.

Instance ID i-0f960cddb16837f (AD_SERVER 1)	VPC ID vpc-04d96053b1bab743e	Security groups sg-060d3ea404da76a29 (Launch-wizard-6)	IAM role -
---	--	--	----------------------

SSM Session Manager | **RDP client** | EC2 serial console

Record RDP connections
You can now record RDP connections using AWS Systems Manager Just-in-time node access. [Learn more](#)

Instance ID
i-0f960cddb16837f (AD_SERVER 1)

Connection Type

- Connect using RDP client**
Download a file to use with your RDP client and retrieve your password.
- Connect using Fleet Manager**
To connect to the instance using Fleet Manager Browser Desktop, the SSM Agent must be installed and running on the instance. For more information, see [Working with SSM Agent](#)

You can connect to your Windows instance using a remote desktop client of your choice, and by downloading and running the RDP shortcut file below:

[Download remote desktop file](#)

When prompted, connect to your instance using the following username and password:

Public DNS
ec2-100-54-66-126.compute-1.amazonaws.com

Username
Administrator

Password
%RnbBR7y0&TQI<cn*U7j6vZkdfxOHRh

Note: If you've joined your instance to a directory, you can use your directory credentials to connect to your instance.

And ensure the pair key permission is changed inorder to login in

```
(phares@matson:~/Downloads)
$ chmod 400 "splunklab.pem"
```

Now lets go and access our splunk lab on terminal and install splunk.
the command is **ssh -i "pair key file" and then the details of the instance** , this

[illegible]

Splunk Enterprise 10.2.2

Index 500 MB/Day. Sign up and download now. After 60 days you can convert to a perpetual free license or purchase a Splunk Enterprise license to continue using the expanded functionality designed for enterprise-scale deployments.

Choose Your Installation Package

 Windows

 Linux

 Mac OS

Architecture	Operating System	Package Format	Size	Download Now	Copy wget link	More
64-bit	4.x+, 5.x+, 6.x+ kernel Linux distributions	.deb	1269.05 MB	Download Now	Copy wget link	More ▾
		.rpm	1665.31 MB	Download Now	Copy wget link	More ▾
		.tgz	1664.91 MB	Download Now	Copy wget link	More ▾

```
[ec2-user@ip-172-31-85-67 ~]$ wget -O splunk-10.2.2-80b9d638de6-linux-x86_64.tgz "https://download.splunk.com/products/splunk/releases/10.2.2/linux/splunk-10.2.2-80b9d638de6-linux-x86_64.tgz"
2026-04-25 07:19:31 (151 MB/s) --> https://download.splunk.com/products/splunk/releases/10.2.2/linux/splunk-10.2.2-80b9d638de6-linux-x86_64.tgz
Resolving download.splunk.com (download.splunk.com)... 3.167.37.33, 3.167.37.31, 3.167.37.32, 3.167.37.34...
Connecting to download.splunk.com (download.splunk.com):[3.167.37.33]:443... connected.
HTTP request sent, awaiting response... 200 OK
length: 174579877 (1.66) [binary/octet-stream]
Saving to: 'splunk-10.2.2-80b9d638de6-linux-x86_64.tgz'

splunk-10.2.2-80b9d638de6-linux-x86_64.tgz
100%[=====] 1.62G 11.0MB/s in 11s

2026-04-25 07:19:13 (151 MB/s) - 'splunk-10.2.2-80b9d638de6-linux-x86_64.tgz' saved [174579877/174579877]

[ec2-user@ip-172-31-85-67 ~]$
```

```

[ec2-user@ip-172-31-65-67 ~]$ ls
gitlink-10.2.2-0000000000-Linux-amd64.tgz
[ec2-user@ip-172-31-65-67 ~]$ file gitlink-10.2.2-0000000000-Linux-amd64.tgz
gitlink-10.2.2-0000000000-Linux-amd64.tgz: gzip compressed data, last modified: Fri Mar 20 13:58:05 2020, from Unix, original size modulo 2^32 177935904
[ec2-user@ip-172-31-65-67 ~]$ ls

```

```
[ec2-user@ip-172-31-65-67 ~]$ tar -xvzf splunk-10.2.2-80b9063d0ee-linux-amd64.tgz
splunk/
splunk/swifttag/
splunk/swifttag/splunk-Enterprise-primary.swifttag
splunk/opt/
splunk/opt/jemalloc-4k-prof/
splunk/opt/jemalloc-4k-prof/include/
splunk/opt/jemalloc-4k-prof/include/jemalloc/
[ec2-user@ip-172-31-65-67 ~]$ sudo mv splunk /opt/
[ec2-user@ip-172-31-65-67 ~]$
```

now after those step now start the splunk the command:

Here the splunk will ask your to configure username and password that your will use to log in on the web page

sudo /opt/splunk/bin/splunk start --accept-license --answer-yes --no-prompt --seed-passwd admin123 --run-as-root

but if your created a user the command will be: ***sudo /opt/splunk/bin/splunk start --accept-license --answer-yes***

```
[ec2-user@ip-172-31-65-67 ~]$ sudo /opt/splunk/bin/splunk start --accept-license --run-as-root
This appears to be your first time running this version of Splunk.

Splunk software must create an administrator account during startup. Otherwise, you cannot log in.
Create credentials for the administrator account.
Characters do not appear on the screen when you type in credentials.

Please enter an administrator username: Masoka
Password must contain at least:
  * 8 total printable ASCII character(s).
Please enter a new password:
Please confirm new password:
Copying /opt/splunk/etc/openldap/ldap.conf.default to /opt/splunk/etc/openldap/ldap.conf.
writing RSA key
writing RSA key
Moving /opt/splunk/share/splunk/search_mrsparkle/modules.new to /opt/splunk/share/splunk/search_mrsparkle/modules.
Splunk> Needle, Haystack. Found.
Warning: ignoring -extensions option without -extfile
Certificate request self-signature ok
subject=CN = ip-172-31-65-67.ec2.internal, O = SplunkUser
Done
[ OK ]

Waiting for web server at http://127.0.0.1:8080 to be available..... Done

If you get stuck, we're here to help.
Look for answers here: http://docs.splunk.com

The Splunk web interface is at http://ip-172-31-65-67.ec2.internal:8080
```

In order to make reboot it self the command is

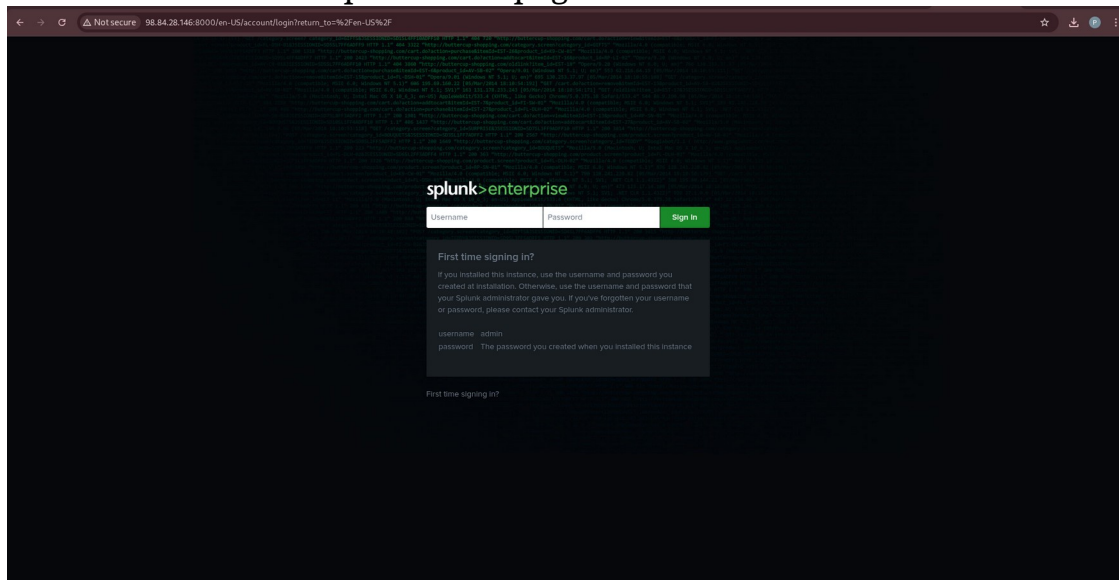
sudo /opt/splunk/bin/splunk enable boot-start

```
[ec2-user@ip-172-31-65-67 ~]$ sudo /opt/splunk/bin/splunk enable boot-start
init script installed at /etc/init.d/splunk.
init script is configured to run at boot.
```

Now will be able to access the splunk web page on our my machine the command is:

Http:// public IP address:8080

and we accessed our splunk web page

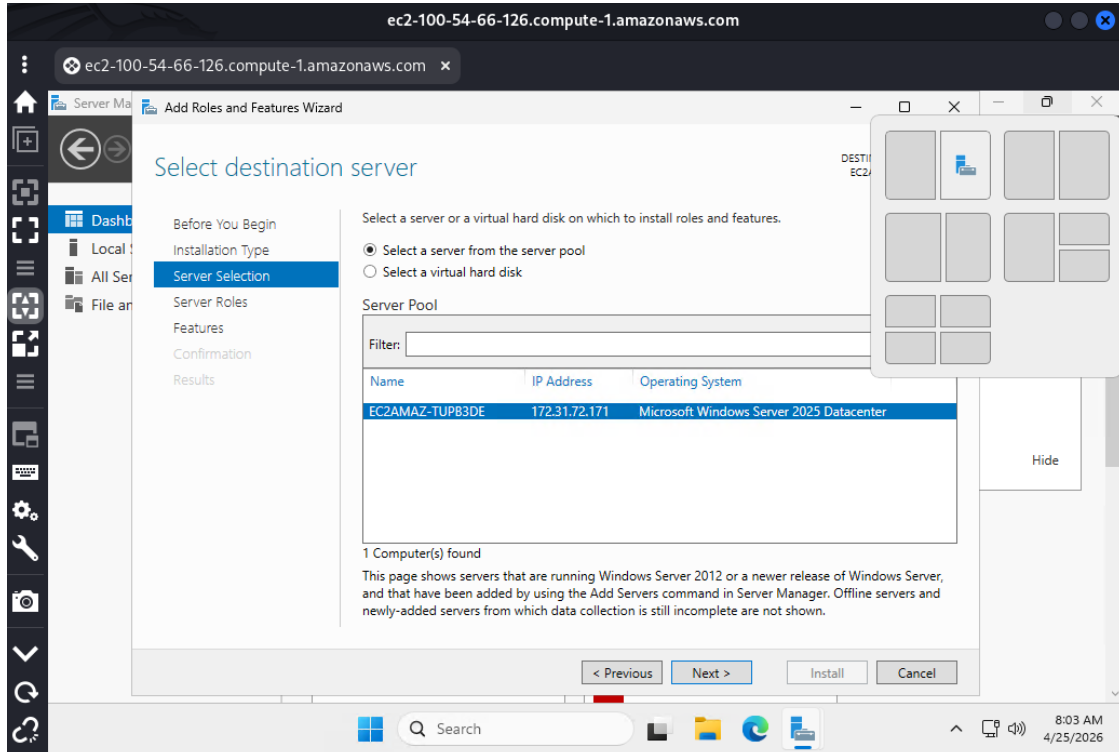
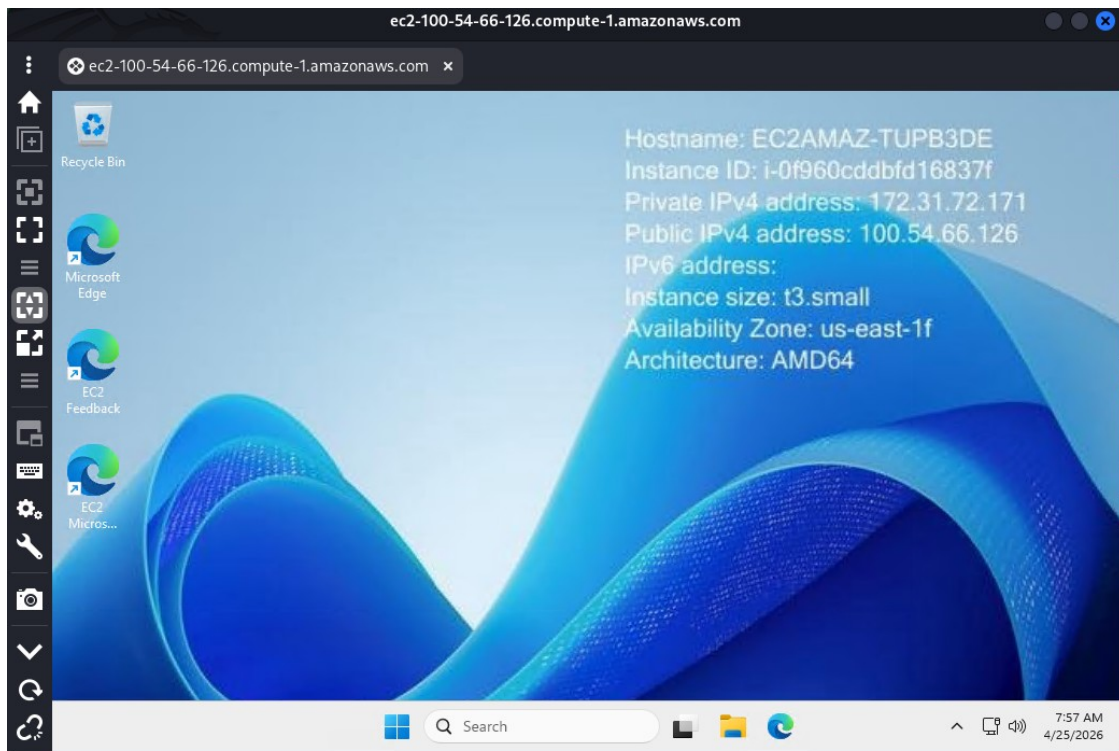


but in order to get this web page you must add a set of inbound rule and security in your splunk server which is TCP and port is 8000 is used for web interface

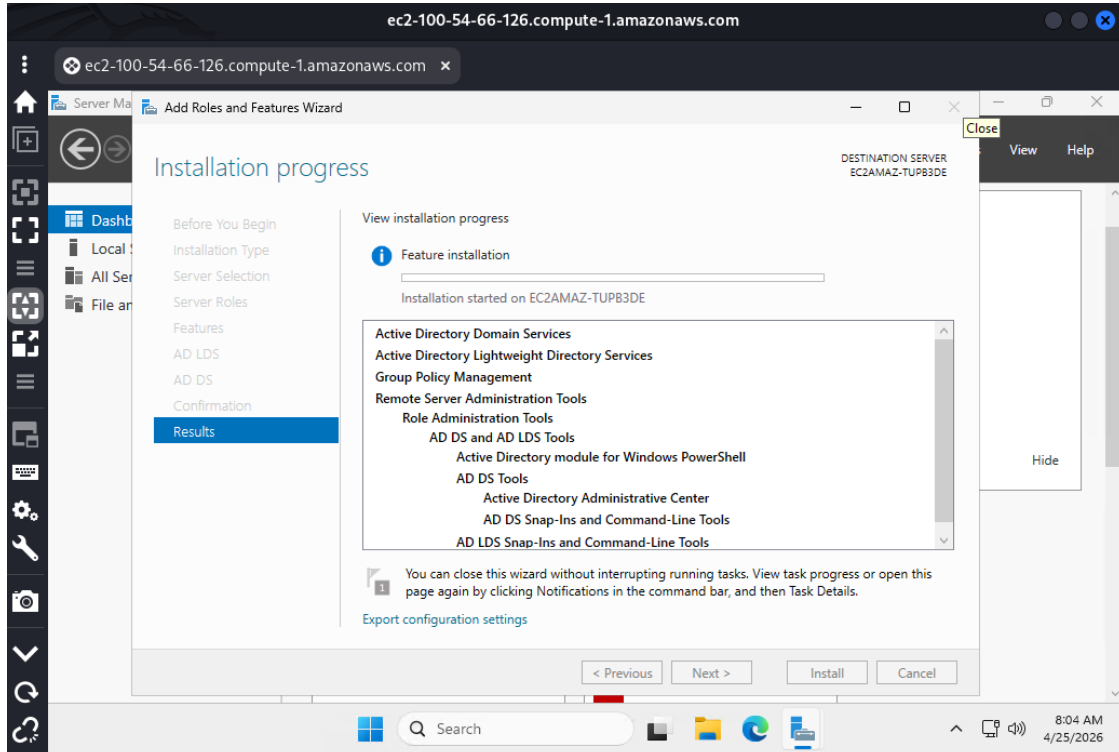
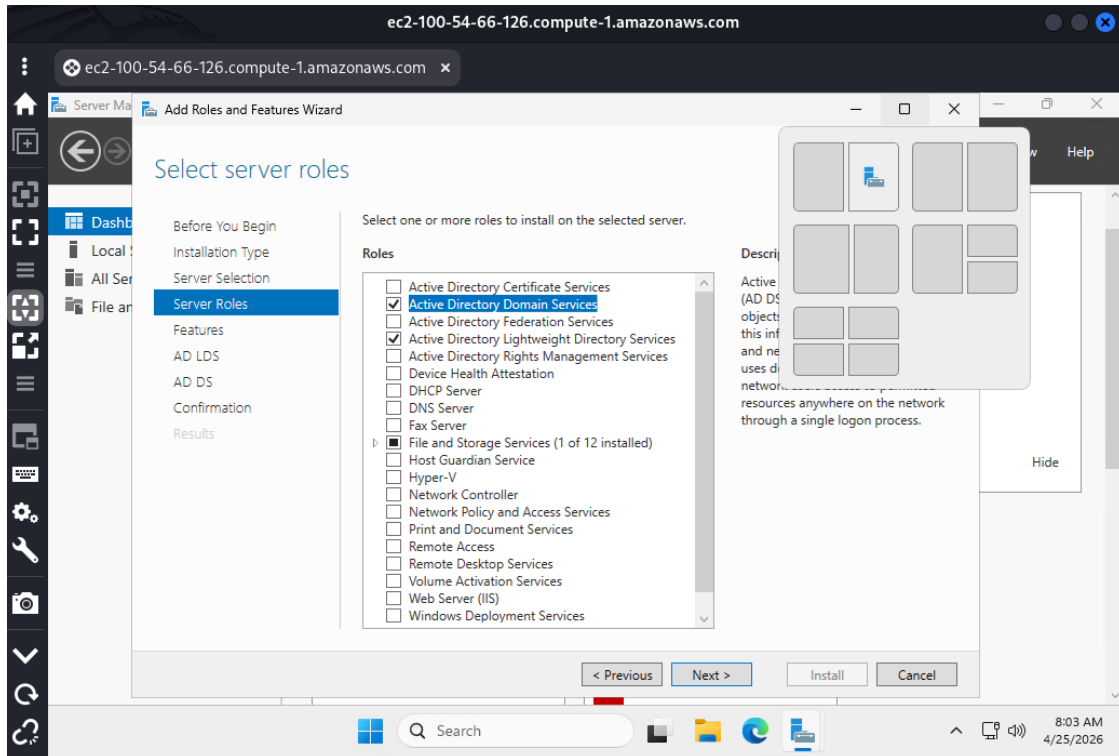
▼ Inbound rules

Name	Security group rule ID	Port range	Protocol	Source	Security groups	Description
-	sg-03c24c47694674f0e	8000	TCP	0.0.0.0/0	splunklab	-
-	sg-0f326a89af663c0fb	22	TCP	0.0.0.0/0	launch-wizard-5	-

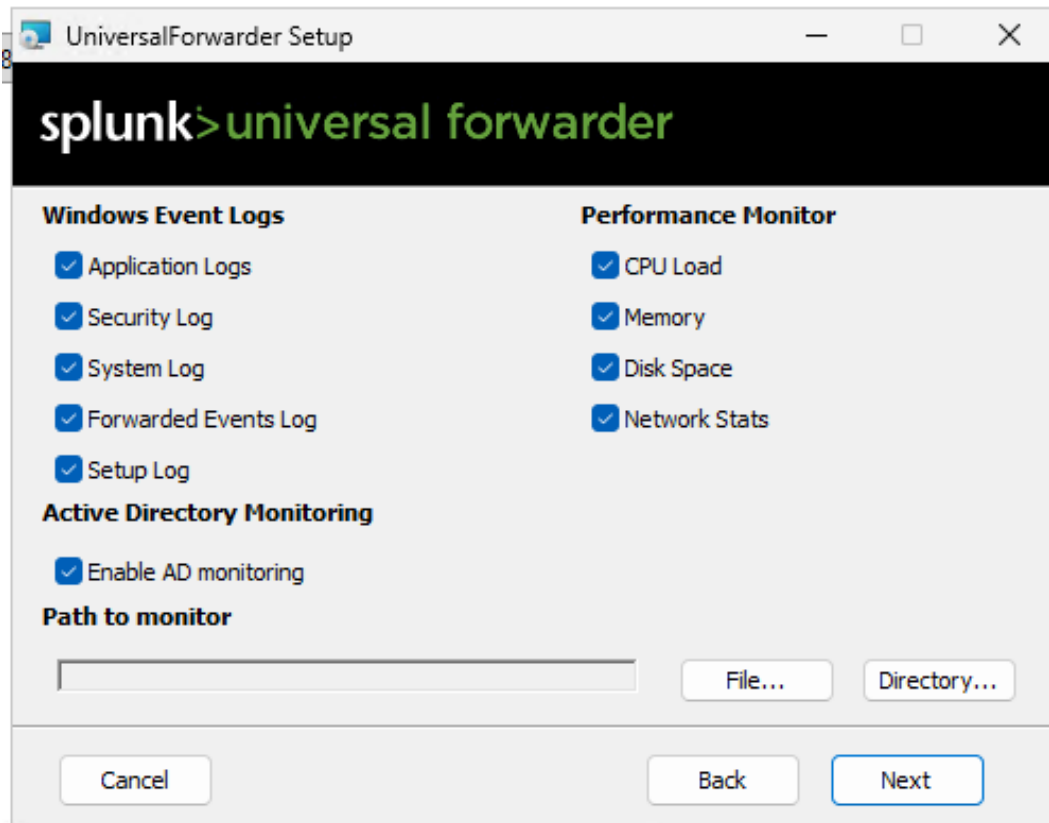
now we got the window server inorder to configure our active directory feature and our universal forwarder in our windows server.



So we need to select AD Domain services and AD lightweight directory services since we will create our forest



Hence the process is complete then you should promote the AD to domain controller so that we will be able to create users, groups and many other things. So after those we will need to install universal forwarder from splunk there are two things to consider which are raw logs you need and the receiving indexer for me I selected all the raw logs I want them.



The Splunk Universal Forwarder was installed on the Windows Server to send logs to the Splunk server.

Configuration included:

- Setting the Splunk server IP as destination
- Using port **9997** for log forwarding

- Selecting relevant log sources

UniversalForwarder Setup

splunk>universal forwarder

If you intend to use a Splunk receiving indexer to configure this UniversalForwarder, please specify the host or IP, and port (default port is 9997). This is an optional step. However, UniversalForwarder needs either a deployment server or receiving indexer in order to do anything.

Receiving Indexer

Hostname or IP

:

Enter the hostname or IP of your receiving indexer, e.g. ds.splunk.com *default is 9997*

also in splunk we need to configure a receiving rule without this we will no get the logs so we will navigate to setting , receiving and forwarding data where we must add port 9997

Receive data

Forwarding and receiving > Receive data

Successfully saved "9997"

Showing 1 of 1 item

Listen on this port	Status	Actions
9997	Enabled Disable	Delete

so we must configure the inbound rule to have this rules below inorder logs to been seen in the splunk web page.

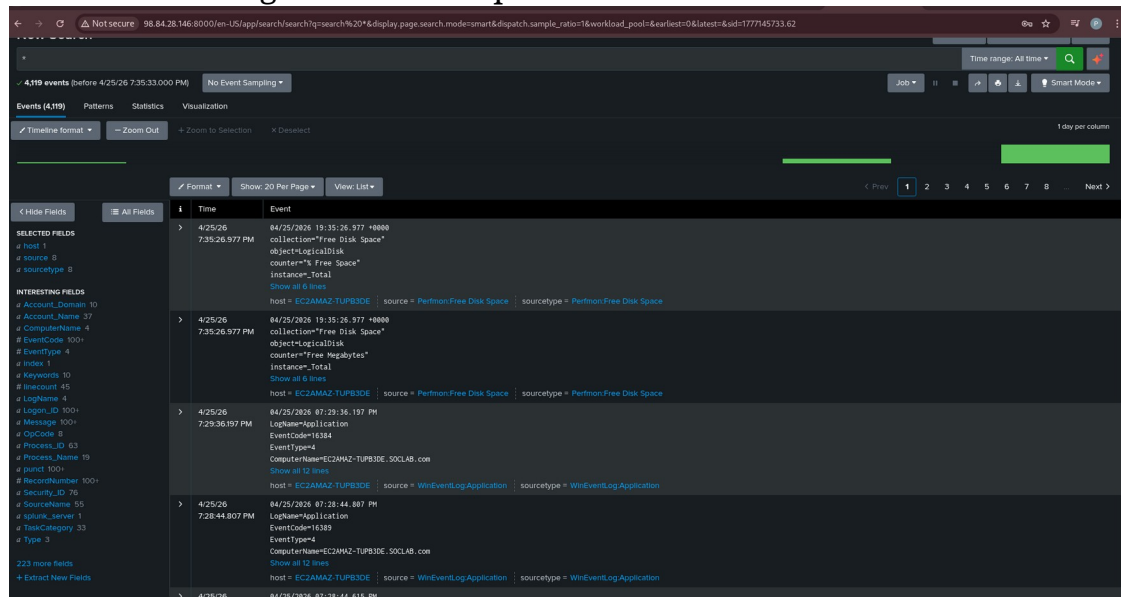
i-Oe03bfefb6d6d831 (SPLUNK LAB)

▼ Inbound rules

Filter rules

Name	Security group rule ID	Port range	Protocol	Source	Security groups	Description
-	sgr-03c24c47694674f0e	8000	TCP	0.0.0.0/0	splunklab	-
-	sgr-02c0e50cb1604a23	9997	TCP	0.0.0.0/0	splunk port	-
-	sgr-0f326a89af663c8fb	22	TCP	0.0.0.0/0	launch-wizard-5	-

and hence the logs seen under splunk tool as the SIEM tool



Log Flow Architecture

The flow of logs in the SOC lab is as follows:

- Windows Server generates logs
- Splunk Universal Forwarder collects logs
- Logs are sent to Splunk Indexer (EC2 server)
- Splunk processes and visualizes logs

This setup enables real-time monitoring and analysis of system activities.

Results

The SOC lab was successfully implemented. The Splunk dashboard displayed logs received from the Windows Server in real time.

Key Achievements:

- Successful deployment of cloud-based SOC lab
- Real-time log collection and forwarding
- Visualization of logs using Splunk dashboards
- Monitoring of user activities and system events

Challenges and Solutions

Challenges:

- Difficulty configuring AWS security group rules

- Initial inability to access Splunk web interface
- Errors during Splunk installation
- Log forwarding issues

Solutions:

- Updated inbound rule sports(8000, 9997)
- Verified correct IP addresses and configurations
- Restarted Splunk services
- Reconfigured forwarder settings

Conclusion

This project successfully demonstrated the design and implementation of a SOC lab using cloud infrastructure. By utilizing **Amazon Web Services** and **Splunk**, a functional environment for monitoring and analyzing security events was achieved.

The project provided valuable hands-on experience in:

- Cloud computing
- SIEM configuration
- Log analysis and monitoring

These skills are essential for a career in cybersecurity, particularly in SOC operations.